

Pretraining Gains Hidden by In-Context Inference

Abstract

Does a low transfer score mean that graph pretraining learned weak representations, or that the model’s inference procedure fails to use them? We distinguish these explanations through stage-resolved evaluation of PRODIGY, comparing native predictions with fixed support-fitted readouts before and after learned in-context inference. On a frozen 500-episode Wiki-to-FB15K237 replication stream, support-centered intermediate ridge improves accuracy by 6.75 percentage points and macro-F1 by 6.83 points. The same pretrained readout exceeds the strongest tested saved-initialization and endpoint-text controls by 8.21 and 8.67 accuracy points, respectively: useful graph-pretraining gains are present but substantially understated by native predictions. The fixed rule also improves fresh-stream social-panel macro-F1 by 2.18–4.13 points under two training schedules. A Cora boundary test separates this finding from a superficially similar outcome: bypassing inference improves classification, yet text-only readouts remain substantially stronger than the graph representations. Post-inference probes and role-matched interventions further show why an intermediate readout’s advantage should not be interpreted as evidence of destroyed information. Together, the results motivate evaluating representation value and inference value separately before deciding whether to change pretraining or deployment.

1. Introduction

Graph in-context learning combines two capabilities: representing graph examples and constructing predictions from a labeled support set. End-to-end transfer evaluation measures their composition. Consequently, the same disappointing score can suggest very different remedies. If representations contain useful task information that native inference fails to exploit, retraining the encoder may address the wrong problem. If those representations are already inferior to the available input features, replacing inference alone is insufficient.

We make this distinction concrete in PRODIGY. On fresh public knowledge-graph episodes, native inference improves accuracy over the stronger of two endpoint-text readouts by only 1.93 percentage points. A fixed support-centered ridge readout on pretrained intermediate features reveals an 8.67-point advantage over the same control. The pretrained readout also exceeds the stronger of two readouts on the exact saved initialization by 8.21 points. Thus the result is neither merely a better classifier on unchanged inputs nor evidence that a random graph encoder suffices. Graph pretraining has produced additional classification utility that native inference leaves partly unrealized.

The location of the readout matters. Applying the same uncentered ridge after metagraph inference performs worse than applying it immediately before inference. However, this does not establish information destruction: labeled support representations and unlabeled query representations have undergone different conditioning. A role-matched intervention partly improves the final-stage probe without closing the gap. The appropriate conclusion concerns the accessibility of representations to a specified support-fitted decision rule, not an architecture-independent measure of all information they contain.

Cora supplies an informative counterpoint. Intermediate readouts improve on native predictions, but both remain substantially worse than text-only classification. This is deployment improvement without evidence of a positive representation advantage over text. Treating every successful bypass as a pretraining success would conflate these cases.

Our contribution is an empirical distinction with a practical consequence. First, we demonstrate substantial, controlled pretraining gains that native inference understates on public relation classification. Second, we establish readout-stage sensitivity and test specific alternative explanations, rather than equating a weak final-layer probe with information loss. Third, we show across social classification and a

public node-classification boundary why representation comparisons and inference comparisons answer different questions. The protocol uses established readout tools; its value is in identifying which part of a transfer pipeline warrants intervention.

2. Stage-resolved evaluation

PRODIGY first encodes sampled graph neighborhoods and pools them into example representations. Its metagraph then propagates information between example and label nodes, using support labels to construct query predictions. We call the last representation before this metagraph **U1**. Native inference uses the complete model; the U1 readout bypasses the metagraph at deployment. A third readout uses the final example representations after metagraph inference.

Within each comparison, we retain the checkpoint, support and query examples, class mapping, and native normalization protocol. The public implementation uses two background layers, endpoint-concatenating pooling, and two metagraph layers (**S2,UX,M2**); social and Cora experiments use **S,U,M**. They belong to the same model family, not an identical architecture configuration. Public replay preserves the native evaluator’s training-mode batch normalization and restores recorded state for paired forwards; social and Cora preserve evaluation mode. Public native-logit parity uses absolute tolerance 10^{-4} and zero relative tolerance, not bit-exact agreement. Saved inputs and outputs are checked against artifact hashes. Cross-implementation results establish scope, not a controlled architecture ablation.

For the centered readout, let support features be the rows of a matrix, and compute their mean. Subtract that mean from both support and query features, then normalize each row to unit length. With the normalized matrices denoted by X_s and X_q , and one-hot support labels by Y_s , we fit

$$W = X_s^\top (X_s X_s^\top + I)^{-1} Y_s, \quad Z_q = X_q W.$$

Regularization and logit scale are both fixed at one, with no intercept. Neither query labels nor the query mean enters the rule. We also report uncentered ridge. Centering was investigated on the discovery stream, then frozen before evaluating a newly constructed public episode stream and the subsequent breadth tests. No target-specific readout tuning is introduced in these applications.

Representation controls use the exact saved initial encoder and a no-GFM feature baseline. Public relation classification concatenates the head and tail text embeddings; node-classification controls use center-node features. These inputs already contain pretrained text information. Our comparisons therefore measure the additional utility of graph-model processing and pretraining, not learning from uninformative inputs.

Accuracy and macro-F1 measure classification decisions; AUC measures ranking; negative log-likelihood (NLL) evaluates assigned probabilities. Public multiclass AUC is computed one-versus-rest within tasks. We report these quantities separately because a readout can improve classification while producing less useful probabilities. Macro-F1 is not inferred from accuracy.

3. Results

3.1. Public pretraining gains exceed native deployment gains

The primary public transfer uses Wiki pretraining and FB15K237 relation classification. The nominated checkpoint, `state_dict_8000`, follows 8,001 optimizer updates under the released loop’s indexing and is fixed before target comparisons. The fresh stream contains 500 twenty-way episodes with three support and four query examples per class. Its sampler differs from the discovery stream while retaining the checkpoint and dataset pools. Under the preserved public split implementation, support and query pools

are disjoint, but entities and examples can recur across episodes. This stream replication is not a second training seed.

Fresh public stream	Accuracy	Macro-F1	OVR AUC
Native inference	73.54	71.94	97.52
Pretrained U1, centered ridge	80.28	78.77	97.83
Saved initialization, uncentered ridge	72.07	69.57	96.13
Saved initialization, centered ridge	69.78	67.43	95.57
Endpoint text, uncentered ridge	71.48	69.72	95.79
Endpoint text, centered ridge	71.61	69.85	95.72

All entries are percentages. The centered pretrained readout improves accuracy over native inference in 463 episodes, loses in 19, and ties in 18. Its accuracy and macro-F1 advantages are 6.75 and 6.83 points. Comparison with both normalization variants of the initialization and text controls prevents the conclusion from depending on a deliberately weak control. The representation advantage persists on the fresh stream after the deployment rule has been fixed.

The probability result is different: NLL increases from 0.873 for native inference to 2.674 for centered ridge at the fixed scale. The public result establishes better classification and a small ranking improvement, not calibration dominance. A deployment requiring reliable probabilities would need a separately validated probability-calibration procedure.

3.2. The readout advantage extends to social classification

We apply the public-fixed readout to native-trained social models under blocked and interleaved schedules. The primary comparison averages four supported target tasks on the fresh episode stream. Existing saved examples and representations are reused, with baseline readouts reproduced before scoring.

Schedule and readout	Accuracy	Macro-F1	AUC
Blocked, native	78.69	78.07	83.04
Blocked, centered U1	80.88	80.25	86.08
Interleaved, native	76.58	75.94	81.62
Interleaved, centered U1	80.66	80.08	84.40

The macro-F1 gains are 2.18 and 4.13 points. Unlike the public twenty-way result, social NLL also improves, from 0.625 to 0.518 for blocked training and from 0.780 to 0.521 for interleaving. Centered raw-feature accuracy averages 73.14%, below both centered U1 means.

Panel-level improvement is not per-target dominance of centering. Relative to uncentered U1, centering helps some target/model combinations and hurts others. These are breadth results from existing social streams and one training seed per schedule, not independent replications for every exported model or episode.

3.3. Cora separates bypass gains from representation gains

The public node-classification boundary uses 128 seven-way, three-shot, four-query Cora episodes per social-pretrained model. This is a custom stratified test pool with transductive graph access, not a standard Cora benchmark score.

Model and readout	Accuracy	Macro-F1	OVR AUC
Blocked, native	40.79	38.37	77.37
Blocked, centered U1	43.89	40.34	76.63
Interleaved, native	31.14	29.26	68.11
Interleaved, centered U1	38.67	35.23	73.50
Raw features, uncentered ridge	57.37	54.79	84.97

Centered U1 improves native accuracy by 3.10 and 7.53 points, yet the raw-feature readout exceeds it by 13.48 and 18.69 points. Native blocked inference also retains higher AUC than its centered U1 replacement. The correct diagnosis is therefore not “useful pretraining hidden by inference.” It is that bypassing inference improves classification while substantial representation disadvantage relative to the inputs remains. Without an initialization control for these transferred models on Cora, that disadvantage cannot be assigned solely to training rather than architecture.

4. What explains—and does not explain—the gap?

The issue is not simply a weak final classifier. On the 500-episode public discovery stream, uncentered pre-metagraph ridge reaches 76.91% accuracy, compared with 70.14% for the same readout after inference and 73.80% for native predictions. Replacing the final classifier without changing the feature stage does not recover the intermediate readout’s utility.

Role conditioning is relevant but not a complete explanation. Support embeddings after inference have been conditioned on their labels; query embeddings have not. In a fixed 64-episode pilot, we hide support labels in additional passes and collect query-role support representations for fitting the post-inference readout. Accuracy rises from 71.27% to 74.12%, but remains below native inference at 75.18% and centered pre-inference ridge at 81.56%. AUC does not recover. The intervention also changes the conditioning set, so it is evidence of partial recoverability, not a complete causal decomposition.

Training/evaluation cardinality mismatch is insufficient. The public training recipe is fifteen-way. A separate 128-episode test at training-matched cardinality still yields a 5.39-point centered-readout accuracy gain and a 5.48-point macro-F1 gain. A twenty-way companion test gives gains of 6.50 and 6.53 points. Examples are not paired across cardinalities, and total batch context changes; these results reject cardinality mismatch as the sole explanation without establishing a cardinality interaction.

A deployment replacement is not automatically a training prescription. A matched social pilot that isolates the encoder from the native inference loss does not improve transfer. This rejects that particular repair, not the general possibility of better objectives or a claim that native inference is necessary for learning. Likewise, an intermediate-stage ranking does not automatically supply a successful cross-target model selector: our tested schedule-selection pilot did not improve held-out performance.

Weak transfer does not imply that observed context is unhelpful. We tested a predeclared alternative: disrupting neighborhood features might improve Cora while harming a target with positive representation transfer. On fixed saved episodes, we permuted real background-node feature rows within each episode while retaining center features, topology, support labels, and the blocked checkpoint. The factual pass reproduced saved representations and native logits before intervention effects were scored. Centered U1 accuracy instead fell on both targets: from 43.89% to 19.34% on Cora and from 86.46% to 60.71% on covid_political. Macro-F1 decreased by 22.41 and 26.34 points, respectively. Thus factual feature–neighborhood associations help relative to scrambled associations even where learned representations trail text. This rejects the proposed opposite-effects explanation, not every form of

harmful aggregation: scrambled context is not a no-context baseline, and the two targets use different episode protocols.

5. Discussion and implications

The practical distinction is between two comparisons: whether pretrained features improve on available input features, and whether native inference exploits those features as effectively as a simple support-fitted alternative. Public relation classification exhibits both positive representation value and an avoidable deployment gap. Cora exhibits a deployment gap without a positive representation advantage over text. Reporting only native transfer—or only the improvement from bypassing it—would obscure this difference.

Simple readouts and centering are established tools, including [SimpleShot](#), [Prototype Rectification](#), and [differentiable ridge regression](#). Our claim is not a new ridge algorithm. It is that graph in-context evaluation should distinguish reusable feature utility from the behavior of a learned, support-conditioned inference procedure. That distinction changes the intervention suggested by a disappointing transfer score.

The evidence concerns one model family, with public relation classification, social classification, and a node-classification boundary. Fresh episodes do not supply independent training seeds, and repeated entities limit episode-level independence. The current results also do not prove that inference destroys information or that bypassing it is universally preferable. They support a narrower, actionable conclusion: before changing graph pretraining to repair transfer, determine whether its gains are absent from the representation or merely absent from the deployed prediction rule.

Work in progress, excluded from the claims above: the second public training initialization is running. Its outcome is not represented as a result in this draft.